

# Generation Interconnection System Impact Study Report

## For Queue Project Q16-0663

Walerga- Antelope 138 kV

May 04, 2018

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0663. The Interconnection Customer proposes a 191.6 MW solar photovoltaic facility interconnecting at the Walerga- Antelope 138 kV substation in the Virginia area. The request was received on August 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 251 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,444,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                    | Contingency                                  | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------|----------------------------------------------|-------------|--------------|----------|
| Intel / South Crk 138.0 kV line       | S Crk / Aibonito Pds 138.0 kV line           | 104.5       | 341          | 7.1      |
| Flowood / Unity Substation 138.0 kV L | Opossum Creek / Parkin 138.0 kV L            | 102.2       | 252          | 7.8      |
| Reach / Cadelaria 138.0 kV Line       | Bear / Clover 345.0 kV line                  | 101.7       | 468          | 6.2      |
| Orlando Coen Lp / Perth 138.0 kV line | Wythe / S Creek 138.0 kV                     | 101.6       | 239          | 5.9      |
| Riv / Perth 138.0 kV                  | S Crk Substation / Wythe Substation 138.0 kV | 100.5       | 236          | 5.9      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                              | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------|----------------------------|---------------|-----------------------|
| Intel / South Creek 138.0 kV Line     | Reconductor line           | \$2,486,000   | 6% (\$151,000)        |
| Flowood / Unity Substation 138.0 kV L | Reconductor line           | \$1,631,000   | 23% (\$379,000)       |
| Reach / Cadelaria 138.0 kV Line       | Reconductor line           | \$4,261,000   | 3% (\$141,000)        |
| Orlando Cogen Lp / Perth 138.0 kV     | Reconductor line           | \$3,457,000   | 16% (\$558,000)       |
| Riv / Perth 138.0 kV                  | Reconductor line           | \$1,330,000   | 16% (\$215,000)       |
|                                       | Total allocated to project |               | \$1,444,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.